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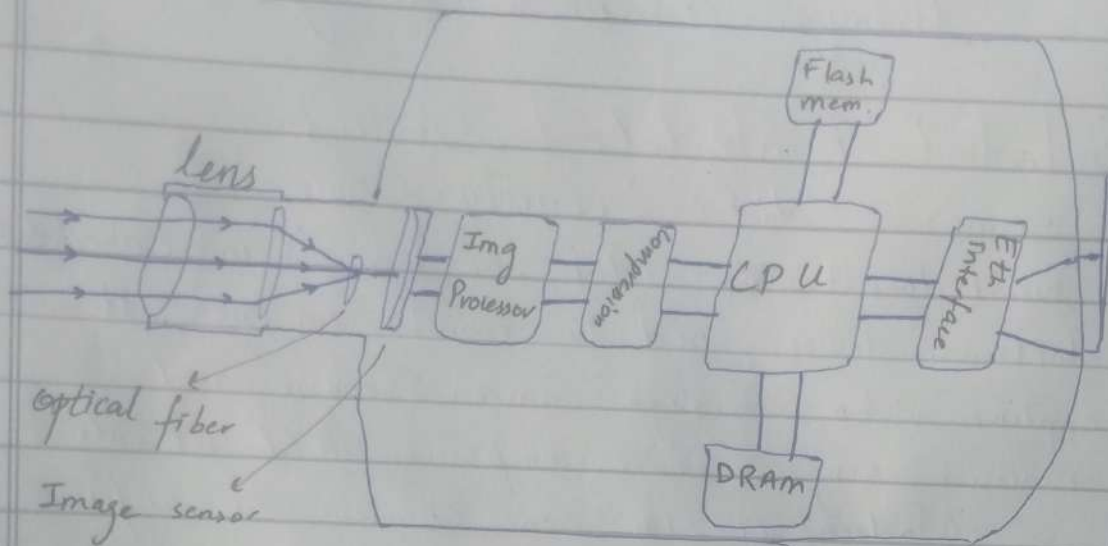
Understanding Computer Evolution through Embedded System

1. Selected Embedded Device:-

- Name: WiFi Security Camera
- What does it do: Wifi sec cam is an embedded electronic device used to monitor an area. It capture video through camera sensor and processes using a small computer built inside.
Modern sec-cam can connect to wifi network and send live video recording to your device.
- Where is it used: It is commonly used in:
Homes, Offices, Shops, Schools, Banks, Parkings etc
- Why does it need computing Tech: Sec cam needs computer technology to process the information collected by its sensors.

Camera needs a processor to convert and process video data. Memory is required while programs are running, storage for saving records. Also, networking hardware to communicate with a router and send video over internet.

2. Basic Computing Components:
 A typical Wi-Fi sec cam contains components that perform functions similar to major parts of a computer.



(wifi sec cam diagram)

- Processor / CPU: The main computing unit of camera. It executes instructions and processes information received from different components.
 In sec cam, it help processing videos, manage OS, handle network, and perform different functions.
- Memory: The camera uses RAM for temp data. For example, when camera is processing live video stream, temp info may be stored in RAM. It is fast but loses memory when power is removed.
- Storage: Storage keeps information for longer period. Storage can be provided by:
 Internal memory, microSD card, Network storage, Cloud.
- Input: Camera sensor acts an input device. It receives light from env and converts it into electric info.
- Output: live video, audio, alerts, notification.
- Power Supply: It provides electric energy to camera's components.

3. Computer History

These machines were developed to perform maths calculation. They are developed with the aim to assist humans with calculation.

Mechanical Calculators.
1600s - 1800s

1800s
Charles Babbage
Analytical Engine.
This was an early concept for programmable mechanical computer as originally planned, its design contained similar ideas to modern computer.

Large Computer like ENIAC used electric components and could perform faster calculation. They were large.

First Electric Computer

1940s
Transistors
It replaced bulky vacuum tubes. Computer becomes smaller, reliable and more efficient.

Now, many Electronic components can be placed together on a single IC. This further reduced size and costs of computer.

1960s
Microprocessors
Placed the main processing functions of a computer onto a small chip.

Computer becomes smaller, affordable, allowing individuals, schools and businesses to use them independently.

1970s - 1980s
Micro Processors
1970s, 1980s
Mobile and Embedded computing.
Computing becomes integrated into mobile phones, vehicles, appliances, industrial equipment and other domains.

Small processors, many wireless communication sensors and efficient software, allows see cars, smart watches, TVs to perform complex tasks. Modern Smart devices.

Today

Q: How has development of computer made today's small and smart devices possible?

Ans: The development of computers has made modern embedded devices possible by making computing smaller, faster, cheaper, and efficient.

→ Smaller processors: Early computers occupied entire rooms while modern processors can fit inside small devices. The transistor, ICs and microprocessors, allowed processing power to be placed into small chips.

→ Memory in smaller devices: Modern memory & storage provides large amounts of data storage in small sizes. Like small SD card can store GBs of data.

→ Wireless communication and networking: The development of networking and wireless technology such as WiFi allows emb. dev. to communicate without requiring a physical connection to a computer.

4. Security Connections

- * Processor:
 - (what it does): executes camera's software and handle tasks such as video processing and dev. mgmt.
 - (what does it matter for sec?): The processor runs the firmware and sec mechanisms of the device. If an attacker exploits a vulnerability, they may be able to make processor execute malicious instructions.
- (what if stop working): The camera may stop operating, becomes unavailable, if an attacker gains control of device, they may potentially access camera function or use the compromised device as part of attack.

(* Storage:

→ What does it do:

Keeps recorded videos configuration information, firmware and other data.

→ Why does it matter for security:

Security camera recording can contain sensitive information. Unauthorized access to stored recording could cause a privacy and confidential problems.

→ What if it stops working:

The camera may fail to record video or previously footage may become unavailable, if an attacker gains unauthorized access to the storage, they may view, modify or delete recorded information.

* Network interface:

→ What it does:

It allows camera to communicate with a router, other devices and other internet services.

→ What does it matter for us?

The network connection creates a communication path b/w camera and other systems.

Weak passcodes, insecure protocol, outdated firmware, or network vulnerability could allow unauthorized access.

→ What happens if it stops working?

The camera may lose remote access or stop sending video and alerts.

If network interface is compromised an attacker could potentially access the camera, or use it as entry point to other systems on the network.